

What factors are associated with motor vehicle risk across the United States? How strong or weak are these associations? What insight did you gain from your analysis?

Introduction.

Traffic accidents have occurred on roads all over the world ever since the invention of automobiles- with the first accident in the United States occurring in the late 19th Century. These accidents are undesirable as they lead to injuries and death of people. Because of this, most governments around the world have created departments focused on ensuring safety on roads through conducting studies on traffic related matters and enacting the findings and recommendations of the reports. These departments also inform the public of their motor vehicle risks through the periodic studies. In the United States, National Traffic Safety Administration (NHTSA) is the body responsible for this function.

This project technically mirrored one of the roles of NHTSA that is- investigating the motor vehicle risk across the States, and giving a judicious judgement based in the findings.

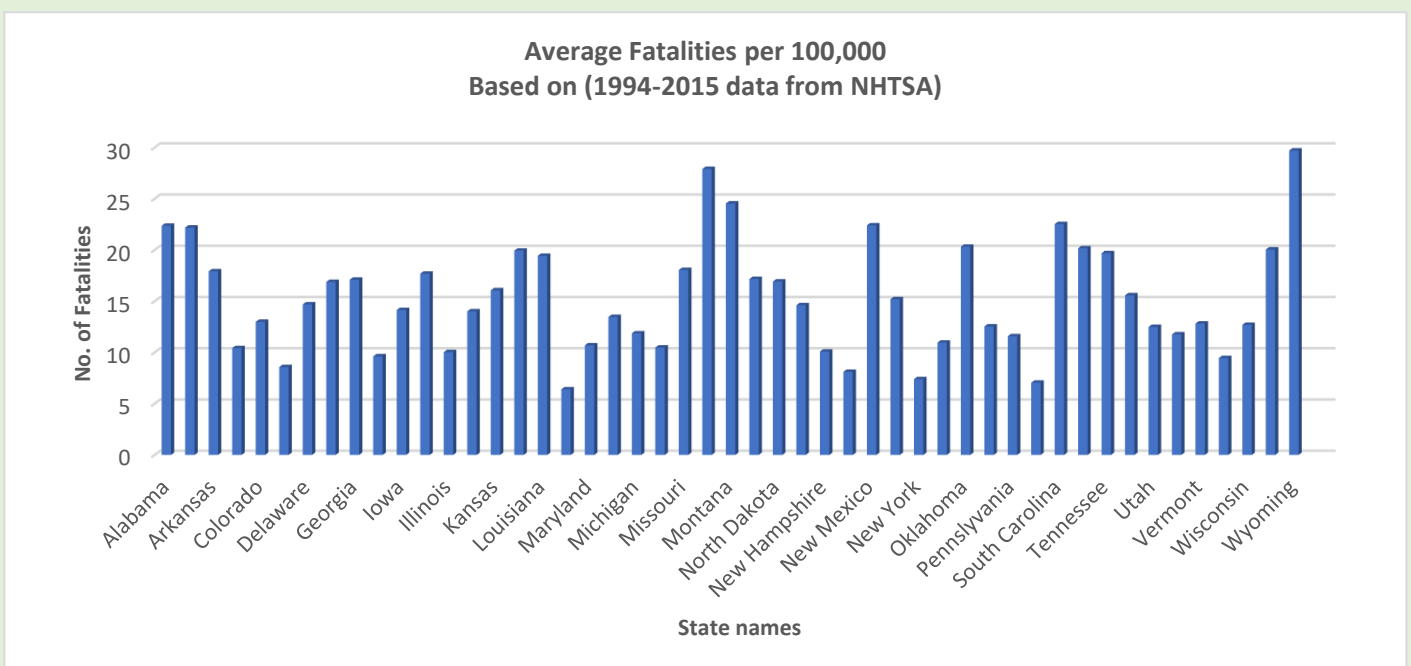
Sources of data and how the data was used.

The measure of Motor vehicle risk identified -Response Variable (RV).

The measure of motor vehicle risk (RV) used throughout this report was generated after a careful consideration of the possible measures of motor vehicle risk found in a Spreadsheet workbook sourced from the NHTSA FARS database. The workbook contained traffic fatalities, resident population, licenced drivers and vehicle miles travelled at each state for the years 1994- 2015. The project generated and settled upon fatalities *per 100,000 people* as the measure of motor vehicle risk. Please note that this response variable- fatalities per 100,000 people, is generated from data collected from 1994-2015 and used to investigate the Motor vehicle risk for 2023. We acknowledge the fact that the exclusion of data for years 2016-2022 could have resulted in less accurate response variable values.

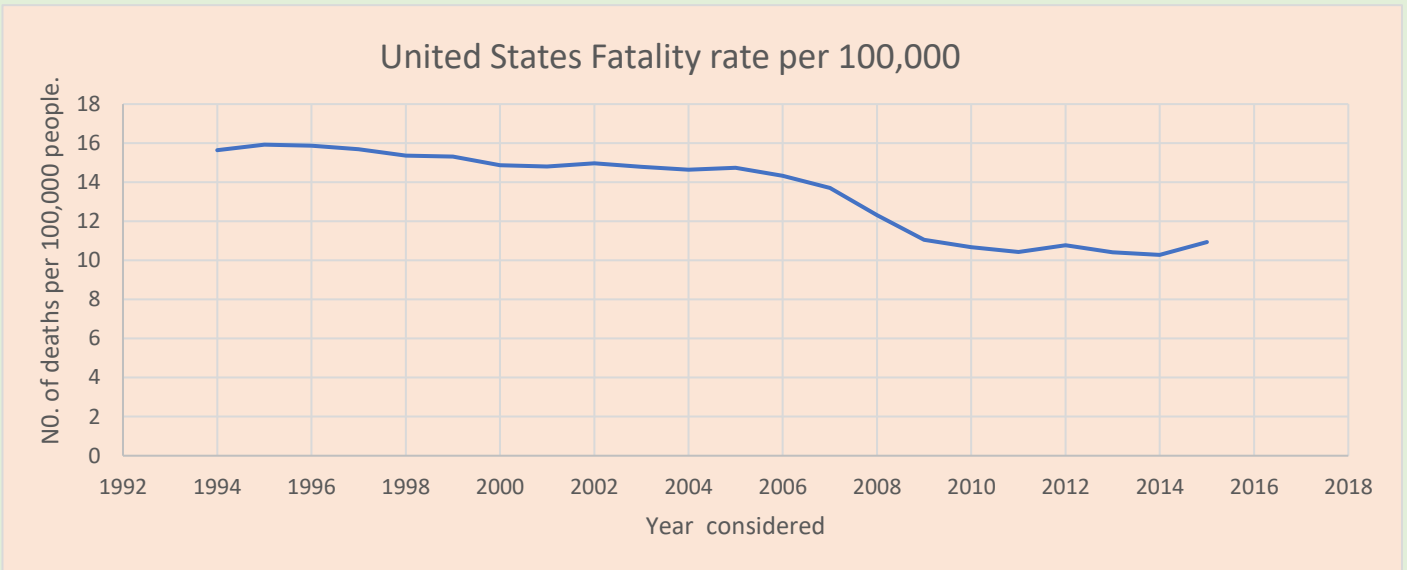
This measure of motor vehicle risk, unlike other possible measures like: Fatalities per vehicle miles (100 million for example) and fatalities per a given number of licensed drivers, closely focuses on the entire population of the state and gives the fatalities for a segment – 100,000 people in this case. This brings about a compelling sense of involvement to everyone in the state, and that's why the project made a strategic decision to use it as a variable to measure the motor vehicle risk.

The population of 100, 000 was specifically chosen since it generated a non-decimal, easy to comprehend rate. The graph below shows the generated fatalities per 100,000 people by State from 1994-2015.



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Using the same Data sourced from the same NHTSA, Fatality rate for the entire United States in each year since the 1994 to 2015 was generated. This is depicted by the graph shown below.



It is important to note that the above graph depicts a reducing overall fatality rate in the country for the time period considered. In 1994 this rate was about 15, however in 2014, it got to 10 fatalities per 100,000 people. The graph clearly confirms the existence of fatalities- which means there is a risk of anyone dying on the US Roads. The existence of this risk makes it logical to go ahead and identify the factors associated with the risk and how they influence the possibility of dying on US roads. These factors are referred to as Explanatory Variables (EVs) on this report.

Explanatory Variables.

As explained before, these are factors which could have an influence on the fatalities rate (RV) already shown in the above graph. This report shall consider the following three EVs:

1. Average Miles per driver in 2022 for each State.

There is a chance that the longer or the shorter a driver drives influences the fatalities rate. This variable was sourced from The Zebra; a top insurance quote company in the US. The company generated these millages using data they collected from the Federal Highway Administration. Their goal was to calculate the average milage per driver across the US and use the values to inform their car insurance rates.

2. Number of full time State and Local Police in each State 2020.

The Number of these officers could influence the fatalities as the project investigated. This variable was collected from Statista. According to European University Institute, Statista is an online portal providing global data on a vast array of sectors for example political data, industrial data and financial/economic data. Statista's main goal is to inform and educate the public.

3. Alcohol consumption per Capita in Gallons 2022 for each State

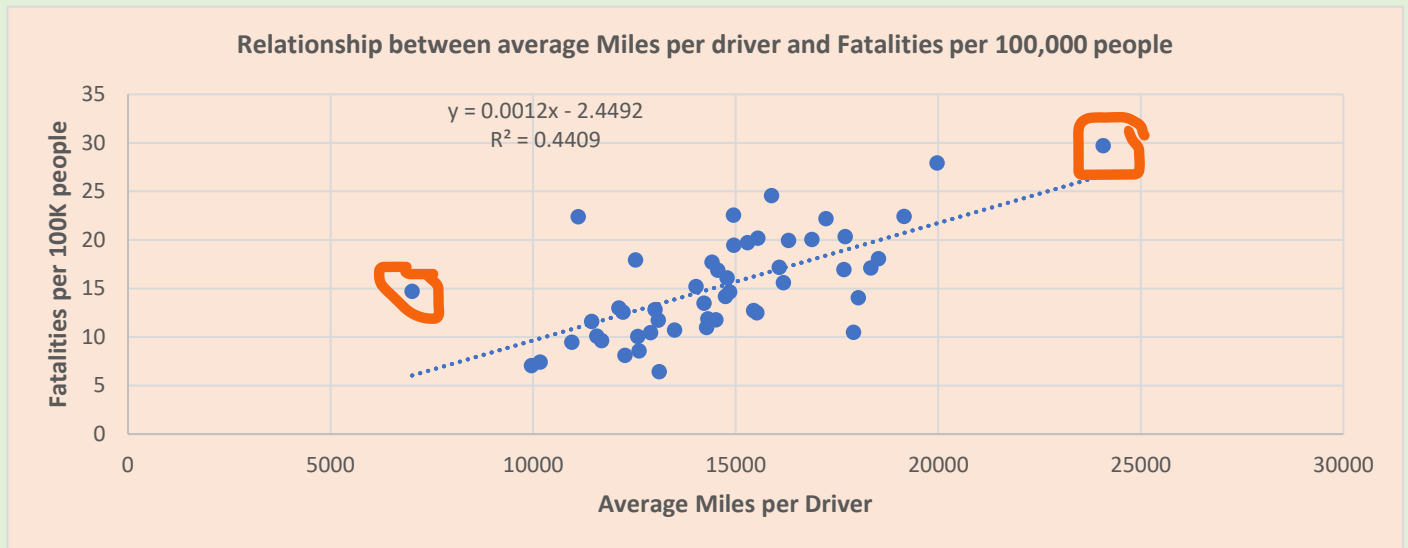
This too could influence the fatalities rate therefore informing the motor vehicle risk. The data was collected from world population review website. A website dedicated to "make data more accessible through graphs, charts, analysis and visualisations". Like Statista, this platform is geared to accurately, efficiently and sufficiently inform the public. However, it is not clear where world population review found their data. Since we do not know their source of data we acknowledge the fact that their sources could be deceitful. Please bear this in mind while reading this report.

Notice that the three EVs whose sources are discussed above are dated 2022, 2020 and 2022 respectively. Since these variables contain data from just a max of two past years, this report assumes that they are highly reliable therefore can be juxtaposed with the Response Variable and correlation Studied.

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Relationship Between Explanatory variables (EVs) and the Response Variable (RV)

1) Average Miles per Driver and Fatalities per 100,000 people.



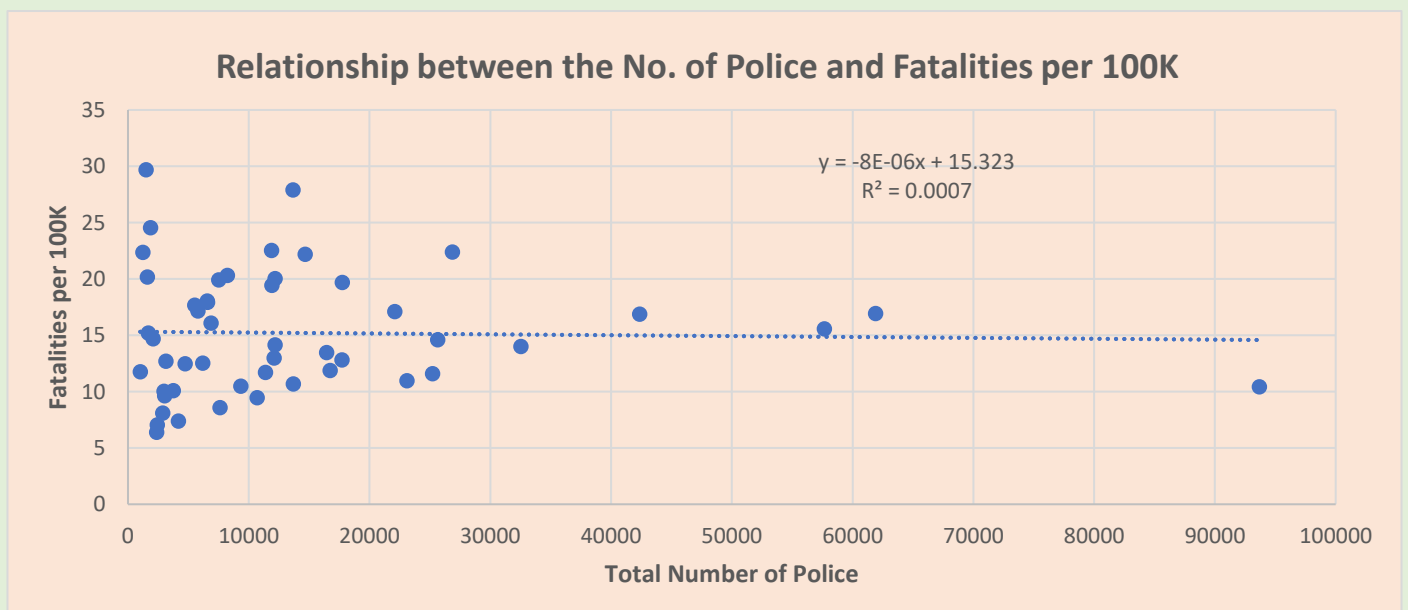
There is a medium-strong positive relationship between the average miles per driver in each State and the resulting value of fatalities per 100,000 people for that specific State. The two variables, (Average Miles per driver and Fatalities per 100 thousand people) have a correlation of 0.66. This correlation confirms the medium-strong positive relationship between the variables. The point to be taken from this analysis is that the average number of miles covered by a driver in any State of the 50 States strongly and positively determines occurrences of fatalities.

The following equation presents a way to calculate the number of fatalities per 100,000 associated with average miles per driver.

$$\text{Fatalities per 100,000} = 0.0012(\text{average miles of Drivers in the State}) - 2.44$$

The bigger the value of miles a driver has on him/herself, the higher the chances of causing/getting involved in a traffic accident. The circled data points on the graphs provide more clarity. Notice that with at an average of about 7000 miles per driver the fatalities hover at about 15 per 100 thousand people. However, at about 24,000 miles per driver the fatalities hovered at 30 per 100 thousand population.

2) Number of Full time State and Local Police and the Fatalities Per 100,000 people.



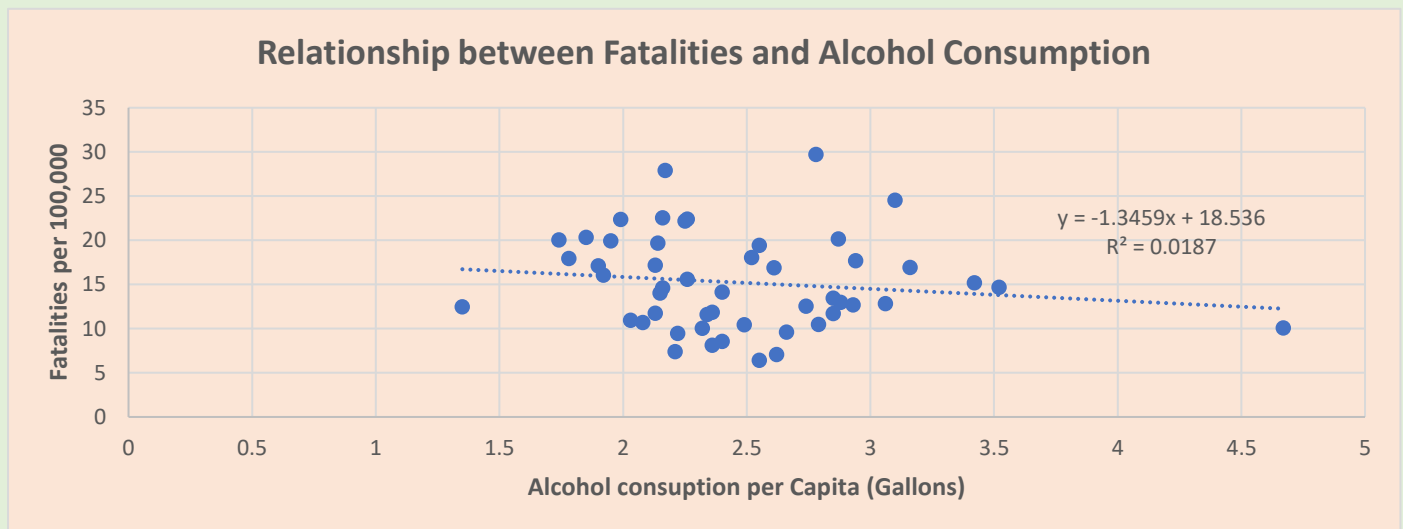
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The graph above shows the relationship between the total number of Police in each of the States and the traffic fatalities per 100 thousand people. The variables have a weak negative correlation of -0.02. This correlation infers that the total number of police (both full time and Local) for a given State, almost doesn't influence the amount of road related fatalities in that State. Number of police has a very minimal negative influence on the number of fatalities per 100,000 people. Negative influence in this case is good, as it means that fewer people die per 100,000. However, as the correlation value confirms, increasing number of police reduces the fatalities at an ignorable slow rate. This is even visually evident on the graph, 15 people die at 10,000 total Police. The same number of people (15) still die at 90,000 total police. It is therefore sensible to claim that Number of employed Police has no effect on the road related fatalities.

The equation below can be used to calculate the number of fatalities per 100,000 based on the Total Number of Police.

$$\text{Fatalities per 100,000} = 8 * 10^{\wedge} - 6(\text{Total Police in State}) + 15$$

3) Relationship between Alcohol consumption (per Capita) and Fatalities per 100,000 people.



Before conducting the data analysis, we hypothesised that Alcohol consumption per Capita would be variable with the Strongest positive correlation with the fatalities. However, it turned out that alcohol consumption in the States had a weak negative correlation of -0.14 to the fatalities. This means that Alcohol has a very minimal influence on fatalities. However, this minimal influence is not an increase in the fatalities rate when more alcohol is consumed but shockingly, a reduction in the number of Fatalities.

On the graph, notice that at about 1.4 gallon per capita 13 people died per 100, 000 people while at about 4.7 Gallons, only ten people died .

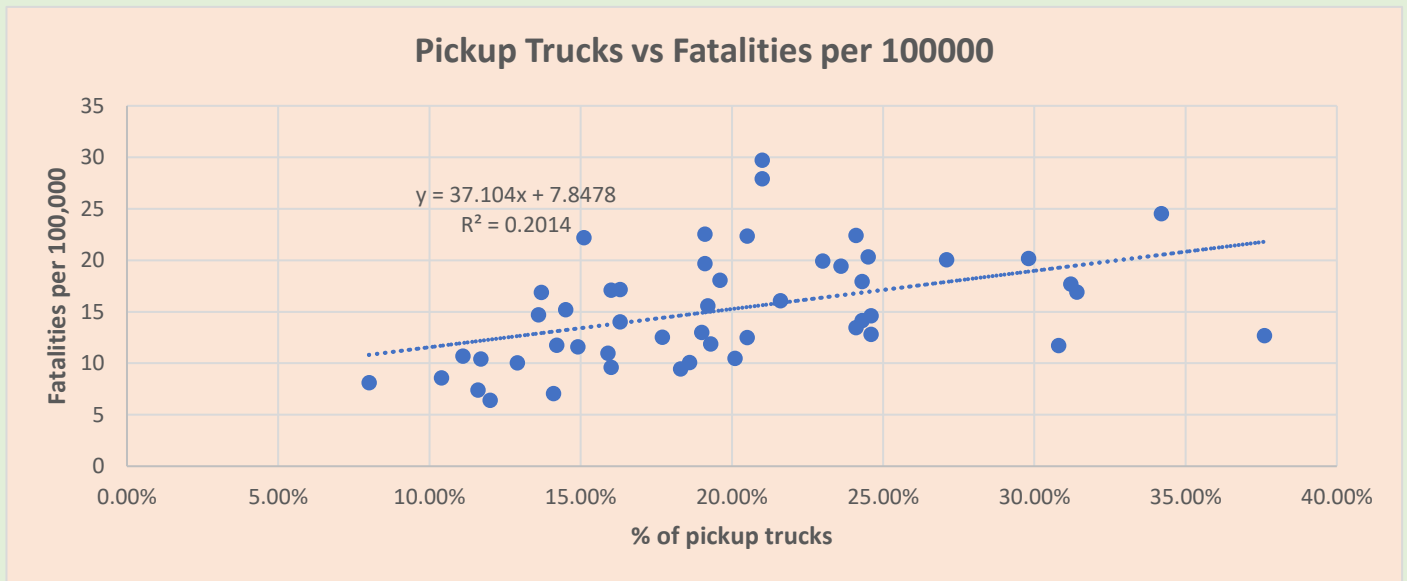
Use the equation below to calculate the number of fatalities based on the alcohol consumption per head.

$$\text{Fatalities per 100,000} = 1.35(\text{Consumption per capita}) + 19$$

According to this study, consuming alcohol reduces the motor vehicle risk; however, the reduction effect is too small that its equivalent to having no influence on Motor vehicle risk. We acknowledge that this realisation is unusual and could have resulted from a deceitful data source mention earlier.

We reiterate that Alcohol consumption (Irrespective of the amount taken) reduces the brain capability to make proper judgement. For this reason do not drive while drunk, as misjudgements could lead to fatal accidents.

4) *Percentage of pickup trucks on the roads for each State vs Fatalities per 100,000.*



With this variable- the percentage of pickup trucks in each State, we found a correlation of 0.4. This is a much stronger correlation compared to the correlations of the two previous variables discussed. This medium strong positive correlation means that the number of trucks in the States is not so strongly but, noticeably responsible for the fatalities in the States .

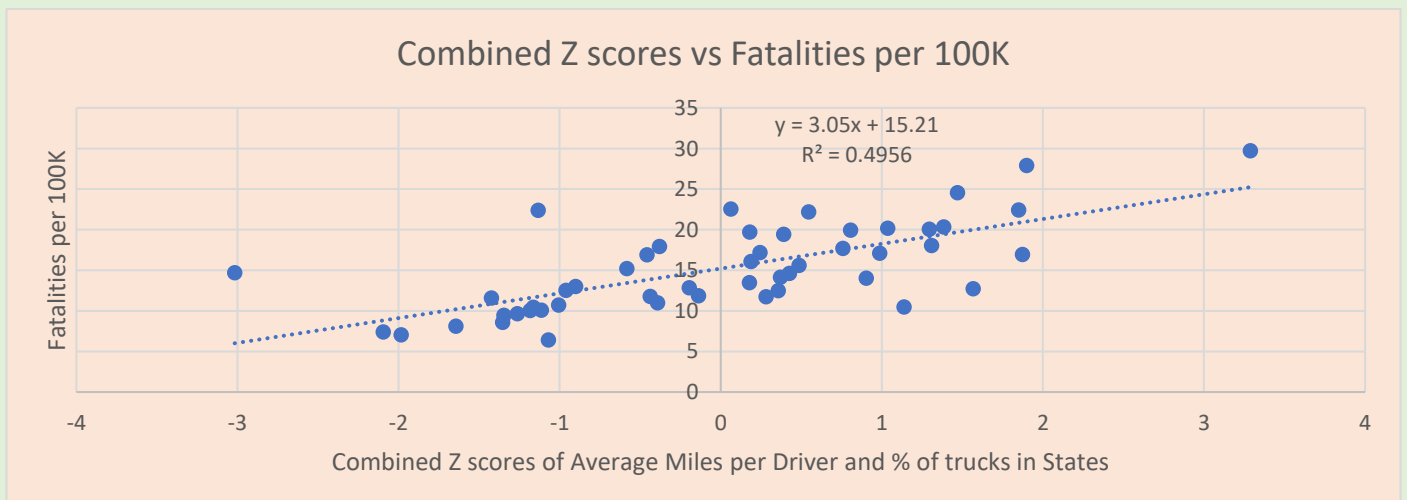
High percentage of pickup trucks on the roads meant a higher number of fatalities per 100, 000 people. The following formula presents a way to calculate the number of fatalities per 100,000 people based on the percentage of pickup trucks on the roads.

$$\text{Fatalities per 100,000} = 37 (\% \text{ of pickup trucks}) + 8$$

Further analysis

Since the average miles per driver and the percentage of trucks on the roads at each state are the variables with the strongest correlation to the fatalities according to our analysis, the panel went a head to generate and combine the Z-scores of the two variables and calculate how the combination of both variables will correlate with fatalities per 100,000 people- the chosen response variable.

1) *Relationship of combined Z-Scores with response variable- Fatalities per 100,000.*



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The combination of Z scores of the two variables most strongly correlated with fatalities (per 100 thousand), resulted in a stronger correlation of 0.7. This is a rise from the individual correlation of 0.44 and 0.66 of two variables earlier. This combination becomes the variable most strongly correlated with the fatalities per 100 thousand people in the Unites.

The strong positive corelation (of 0.7) observed on this combination means that the average number of miles on each driver at a given State combined with the percentage of Pickup drivers in that specific state strongly influence the outcome of fatalities per 100 thousand people. As evident in the equation below, this combination reveals that the more a driver drives and the higher the percentage of pickups trucks in a state, the higher the fatalities per 100 thousand people.

$$\textbf{Fatalities per 100,000} = 3.05 (\textbf{combined Z.Score}) + 15$$

State governments can use this realisation to formulate a policy that would result in reduced traffic accidents. Such policies would focus on reducing the number of milage per driver therefore consequently bringing down the average miles per driver in the state, and also reducing the number of pickup trucks in the State. Reducing the number of pickup trucks would mean low percentage of pickup in that Specific state.

Since these variables strongly influence the fatalities, policies focused on this kind of reduction will essentially reduce the road fatalities per 100 thousand people thereby reducing the motor vehicle risk.

Conclusion

Average miles per driver and the percentage of trucks on roads proved to be the strongest determinants of road related fatalities in the United States. This means that they contribute hugely to the motor vehicle risk compared to the number of Police and Alcohol consumption. From this analysis, a potential solution that can drastically reduce traffic fatality is formulating a regulation around millage on each driver and the percentage of pickup trucks.

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